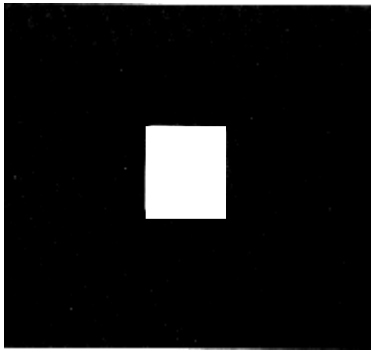
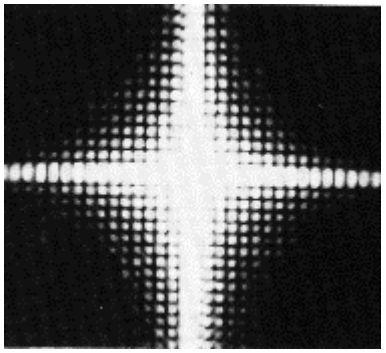
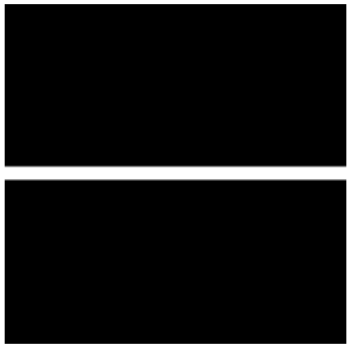
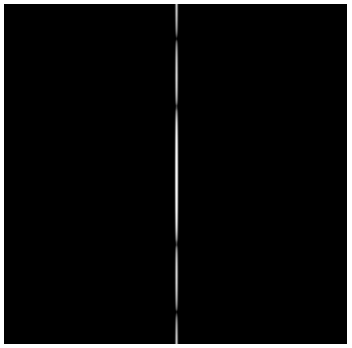
2D Image

Fourier Spectrum

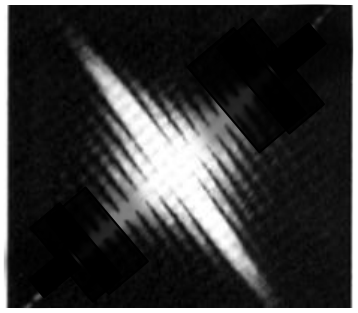












Convolution

Flip g around x
and shift it: $x \rightarrow 0$,
(*then* take the inner product)

1D (continuous, discrete)

$$f * g \quad (x) = \int_{\alpha=-\infty}^{\infty} f(\alpha) g(x - \alpha) d\alpha$$

1D continuous

Good for:
- **Pattern matching**
- **Filtering**
- **Understanding Fourier properties**

$$= \sum_{\alpha=0}^{N-1} f(\alpha) g(x - \alpha)$$

1D discrete

2D (continuous, discrete):

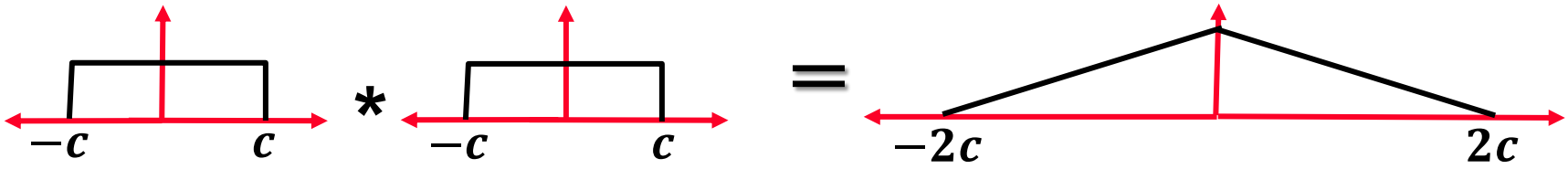
$$f * g \quad (x, y) = \int_{\alpha=-\infty}^{\infty} \int_{\beta=-\infty}^{\infty} f(\alpha, \beta) g(x - \alpha, y - \beta) d\alpha d\beta$$

2D continuous

$$= \sum_{\alpha=0}^{N-1} \sum_{\beta=0}^{N-1} f(\alpha, \beta) g(x - \alpha, y - \beta)$$

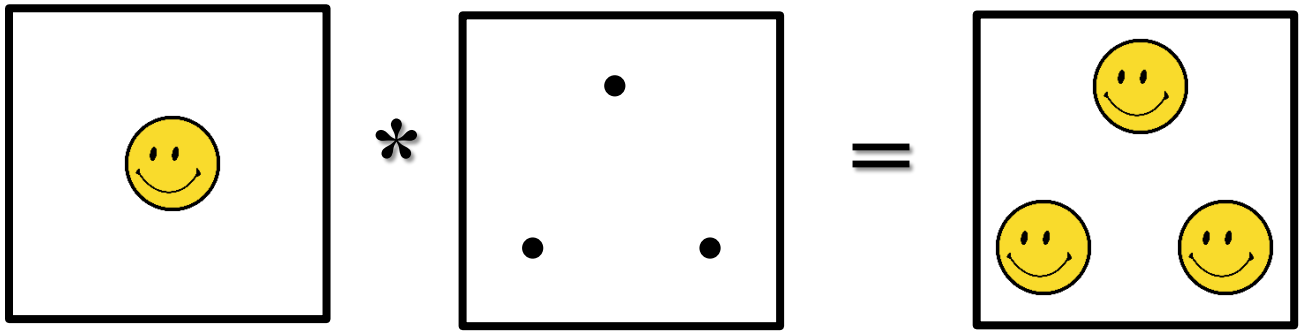
2D discrete

Examples

1) 

2) $f(x) * \delta(x) = f(x)$ **Unity**

3) $f(x) * \delta(x-d) = f(x-d)$ **Shift**

4) 

Replicate

Convolution Properties

- Commutative:

$$f*g = g*f$$

- Associative:

$$(f*g)*h = f*(g*h)$$

- Homogeneous:

$$f*(\lambda g) = \lambda f*g$$

- Additive (Distributive):

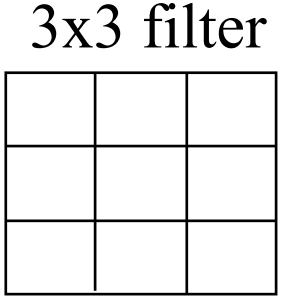
$$f*(g+h) = f*g+f*h$$

- Shift-Invariant

$$f*g(x-x0,y-yo) = (f*g)\,(x-x0,y-yo)$$

Proofs: Homework

Spatial Filtering Operations

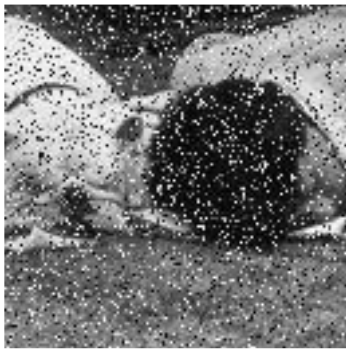


Example 3x3 filter

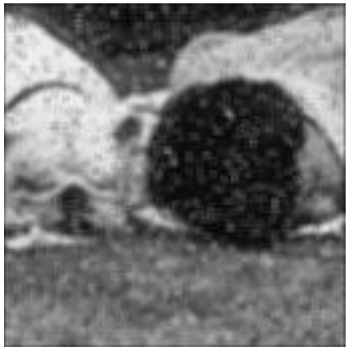
$$h(x,y) = f(x,y) * \frac{1}{9}\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$h(x,y) = 1/9 \sum_{(n,m)} f(n,m)$$

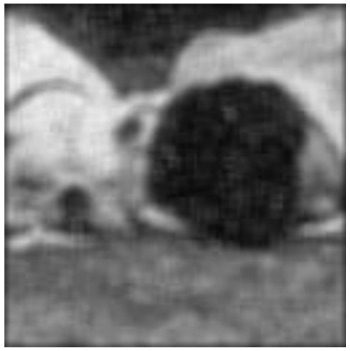
Average of all pixels in the 3x3 neighborhood of (x,y)



Salt & Pepper Noise

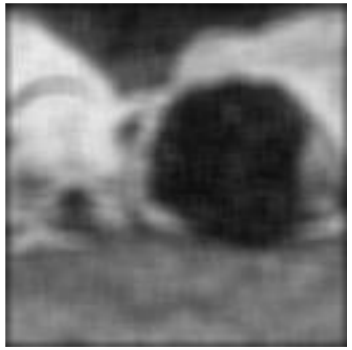


3 X 3 Average



5 X 5 Average

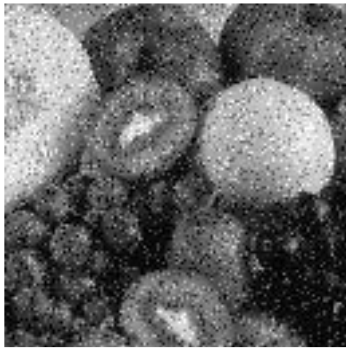
Local averaging:
Removes noise but blurs edges



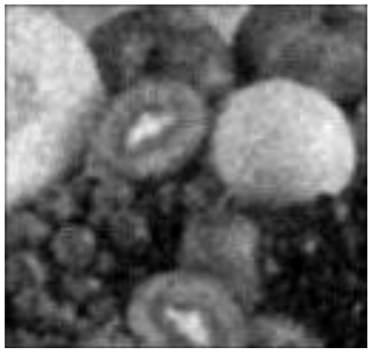
7 X 7 Average



Median



Salt & Pepper Noise

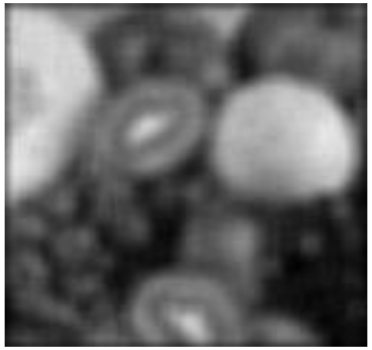


3 X 3 Average

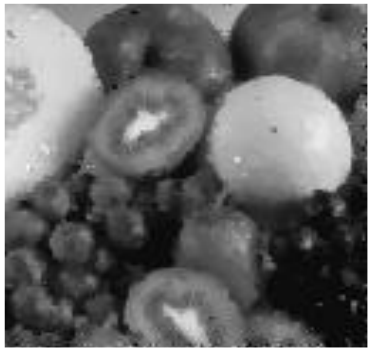


5 X 5 Average

Local averaging:
Removes noise but blurs edges



7 X 7 Average



Median



$$f(x,y) * \begin{bmatrix} 1 & -1 \end{bmatrix}$$

$$= f(x+1,y) - f(x,y)$$

x derivative $\frac{\partial f}{\partial x}$

$$f(x,y) * \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

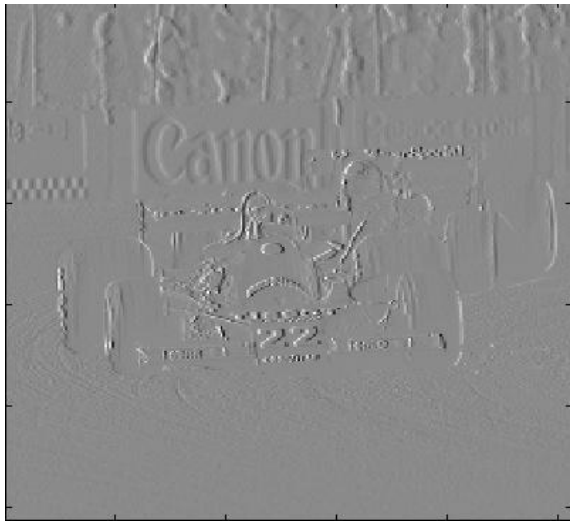
$$= f(x,y+1) - f(x,y)$$

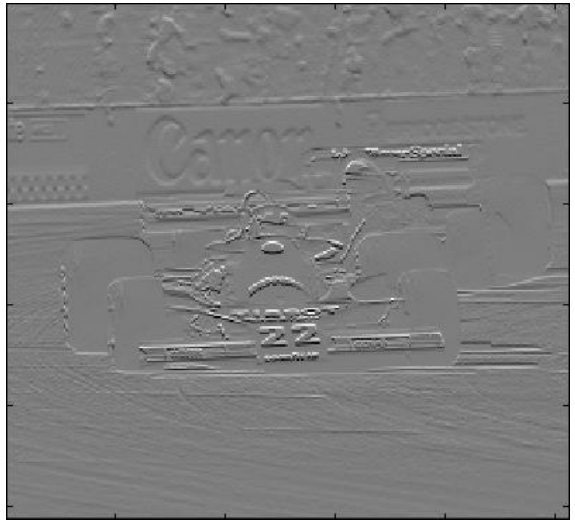
y derivative $\frac{\partial f}{\partial y}$

A very simplistic "Edge Detector"

Gradient magnitude

$$\|\nabla f\| = \sqrt{f_x^{\,2} + f_y^{\,2}}$$







The Convolution Theorem

$$\Im\{f(x) * g(x)\} = \Im\{f(x)\} \cdot \Im\{g(x)\}$$

Proof: Homework

and similarly:

$$\Im\{f(x)g(x)\} = \Im\{f(x)\} * \Im\{g(x)\}$$

➔ Computational complexity of Discrete Convolution:

O(N *log* N)

Take f and g → FFT → $F \cdot G$ → FFT^{-1} → $f * g$

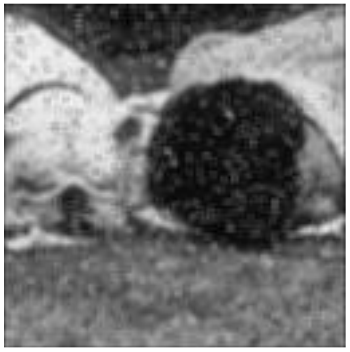
Going back to the "noise cleaning" example...



Salt & Pepper Noise

$$* \quad \frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} =$$

rect

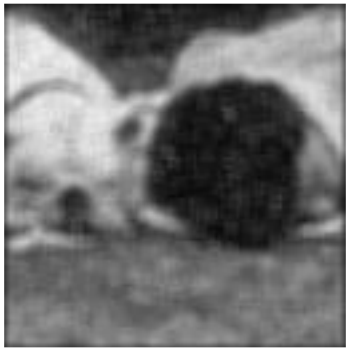


3 X 3 Average

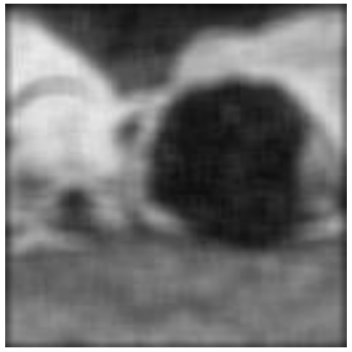
Convolution with a ***rect*** ↔ Multiplication with a ***sinc*** in the Fourier domain

= ***LPF*** (Low-Pass Filter)

➔ **blurs the image**



5 X 5 Average

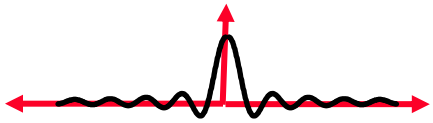
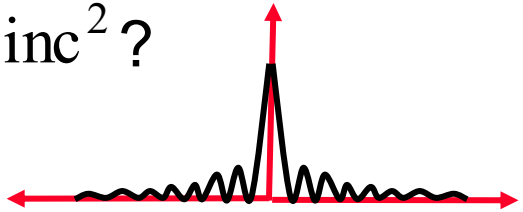


7 X 7 Average

Wider ***rect*** ↔ Narrower ***sinc*** = Stronger ***LPF*** ➔ **Stronger blur**

Examples

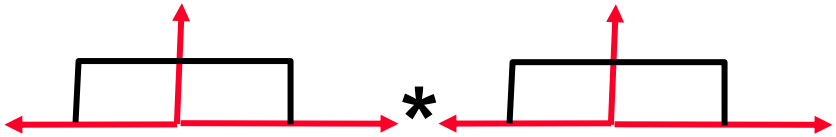
What is the Fourier Transform of $f(x) = \text{sinc}^2$?

$f(x) = ($  $)^2 =$ 

$$\Im\{f(x)\} = \Im\{\text{sinc} \cdot \text{sinc}\}$$

$$= \Im\{\text{sinc}\} * \Im\{\text{sinc}\}$$

$$= rect * rect$$

$=$ 

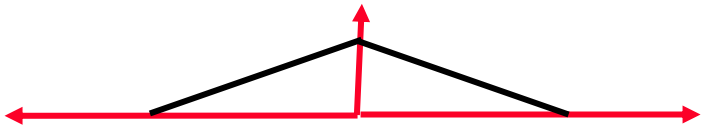
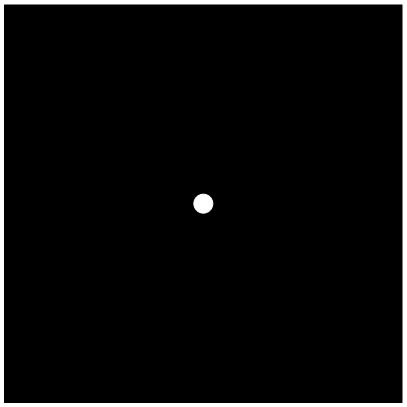
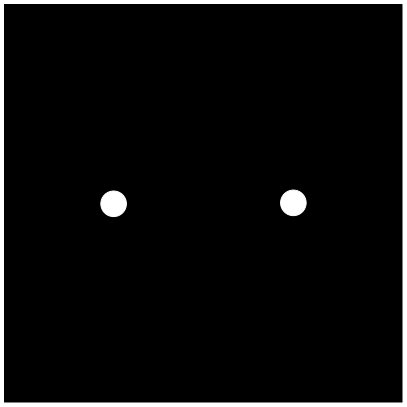
$=$ 

Image Domain

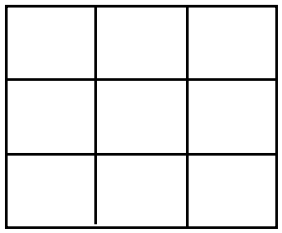




LPF in *Image domain* vs. *Fourier domain*

Computational complexity of convolution: O(N *log*N) via Fourier.
But for filter with a small support *k* ➔ O(*k*N) in the Image domain.

k x ***k*** **rect**



1) $f(x,y) *$ ⟺ F (u,v) ***sinc*** **(u,v)**

Convenient ***But not an ideal LPF...***

(small spatial support;
no cyclic assumption)

2) $f(x,y) * \text{sinc}(x,y)$ ⟺ F (u,v) ***rect*** **(u,v)**

* ***Inconvenient*** to apply spatially (large support) ***An ideal LPF***

* Induces ***"Ringing" artifacts***

3) $f(x,y) * gaussian(x,y)$ ⟺ F (u,v) Gaussian (u,v)

A gaussian = optimal locality in both domains

➔ ***Convenient to apply spatially & Reasonably good LPF...***

The Sampling Theorem

Developed on the board (also found in recording):

- **Nyquist frequency**
- **Aliasing**
- **Ideal interpolation**
- **Discretization induces a Cyclic function**



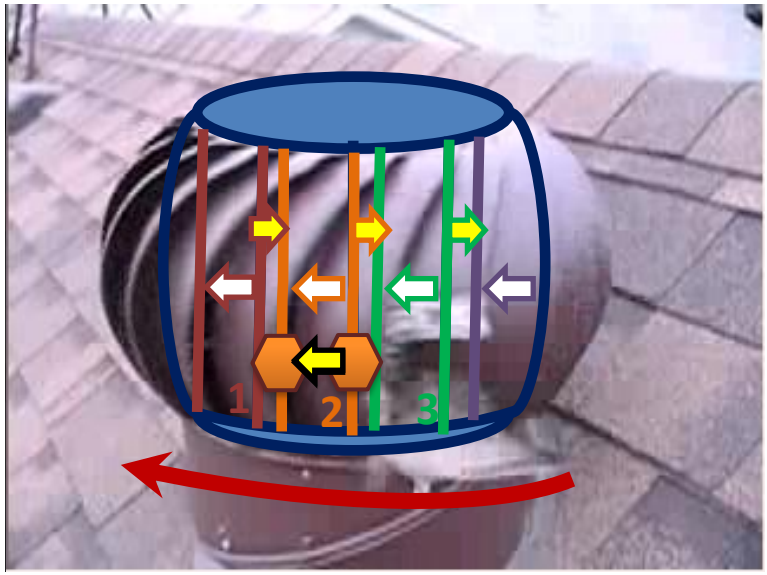
Examples of Temporal Aliasing

A "fast" dynamic event imaged by a "slow camera"

Motion aliasing



Examples of Temporal Aliasing



What's going on…?!

- The turbine rotates fast to the left.
- Each blade moves almost to the next blade position between frames.

➔ So, we wrongly perceive rotation of the blades to the right

➔ ***TEMPORAL ALIASING!***

It happens because the temporal sampling rate (framerate) is too slow.

Multi-Scale Image Representation

- Gaussian pyramids
- Laplacian Pyramids
- Wavelet Pyramids



Image Pyramid

Low resolution



High resolution

Good for:

- pattern matching
- motion analysis
- image compression
- other applications

Fast Pattern Matching

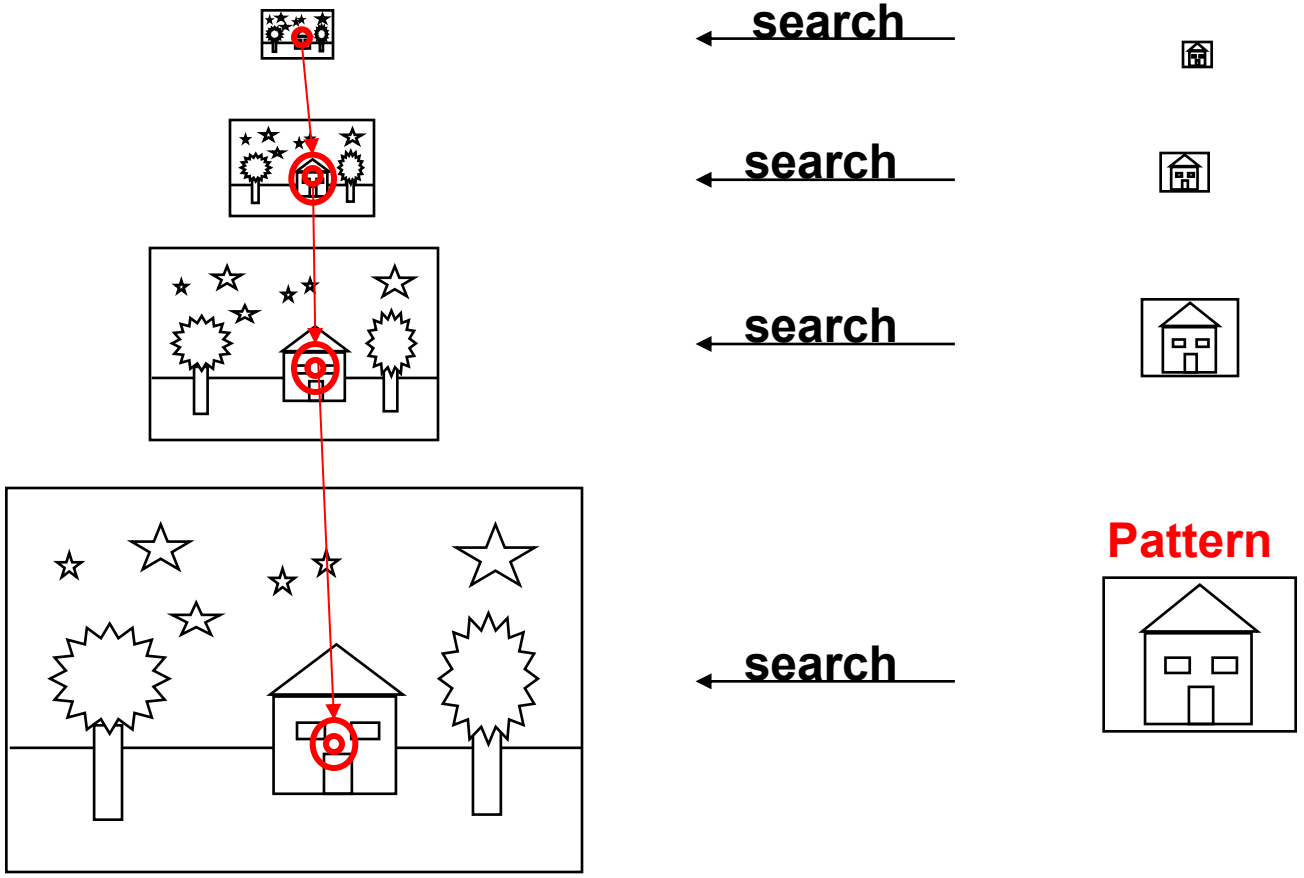


Image Pyramid

Low resolution



High resolution

<u>Does not consume much space</u>

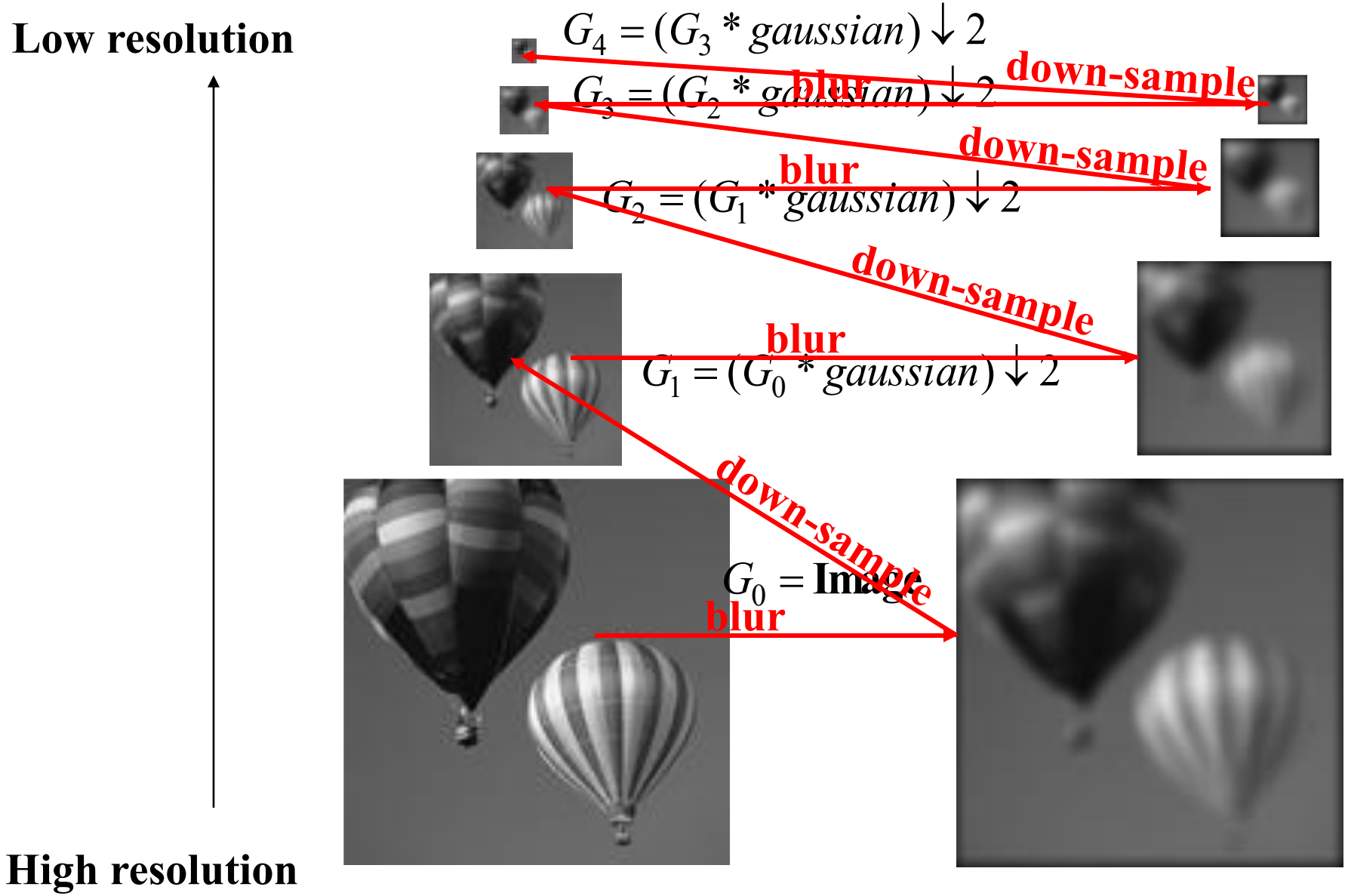
For an NxM image:

$$NM + \frac{1}{4}NM + \frac{1}{16}NM + \ldots$$

$$= \sum_l (\frac{1}{4})^l \, NM$$

$$\leq 1\frac{1}{3} \, NM$$

The Gaussian Pyramid



The Laplacian Pyramid

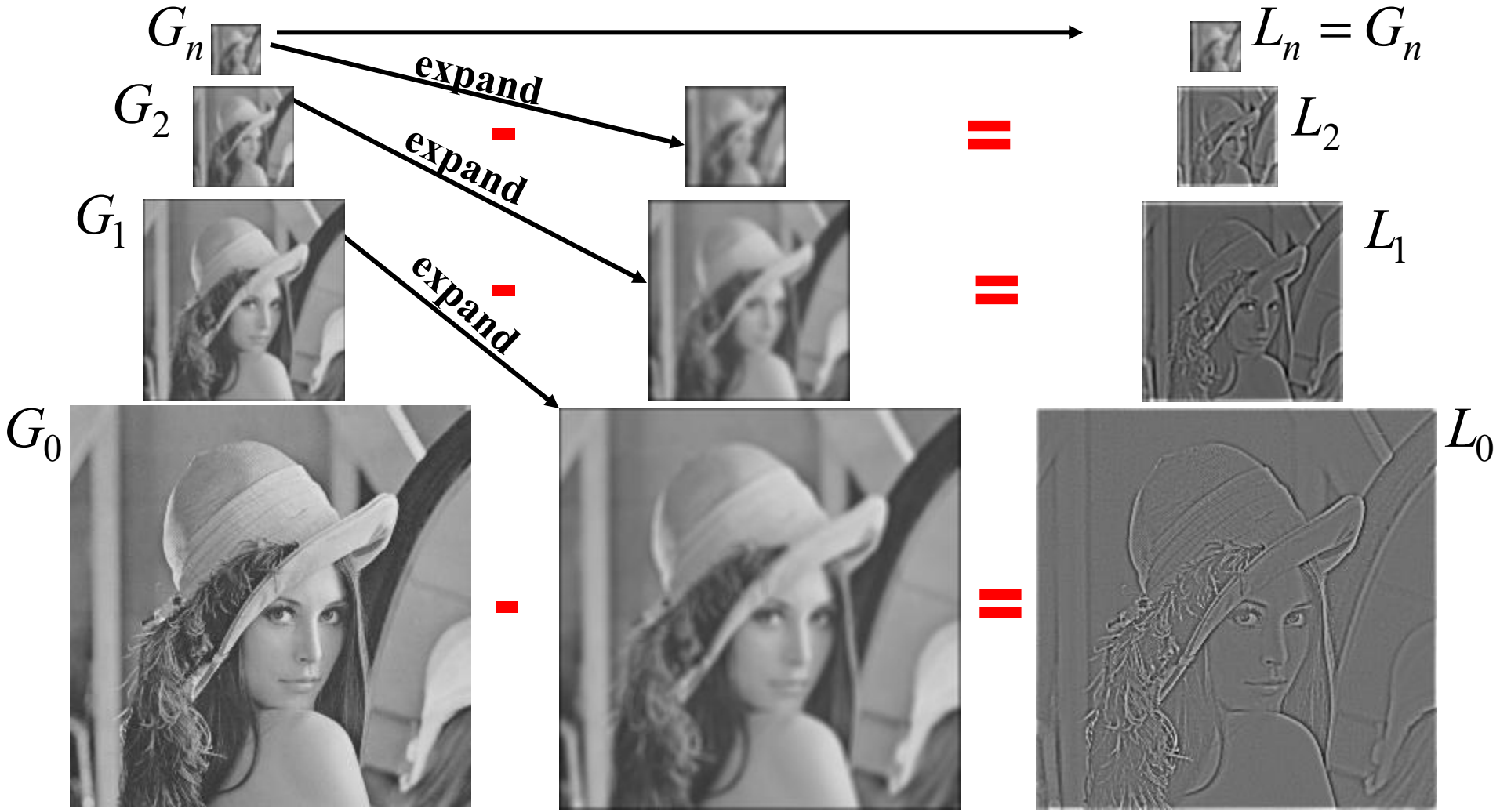
$$L_i = G_i - \text{expand}(G_{i+1})$$

Useful for image compression

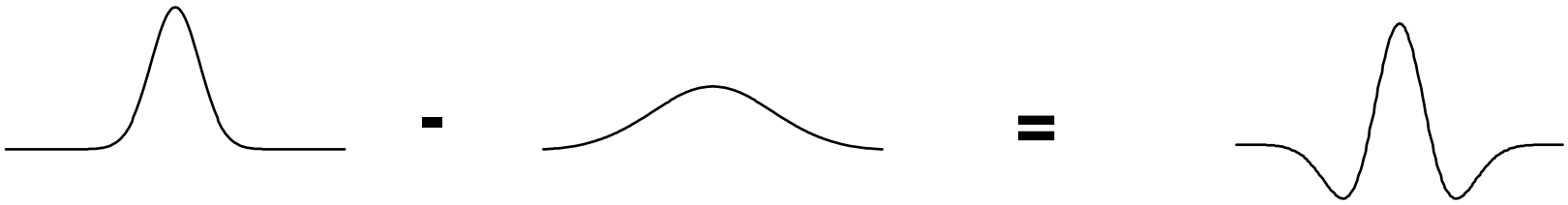
Gaussian Pyramid

$$G_i = L_i + \text{expand}(G_{i+1})$$

Laplacian Pyramid



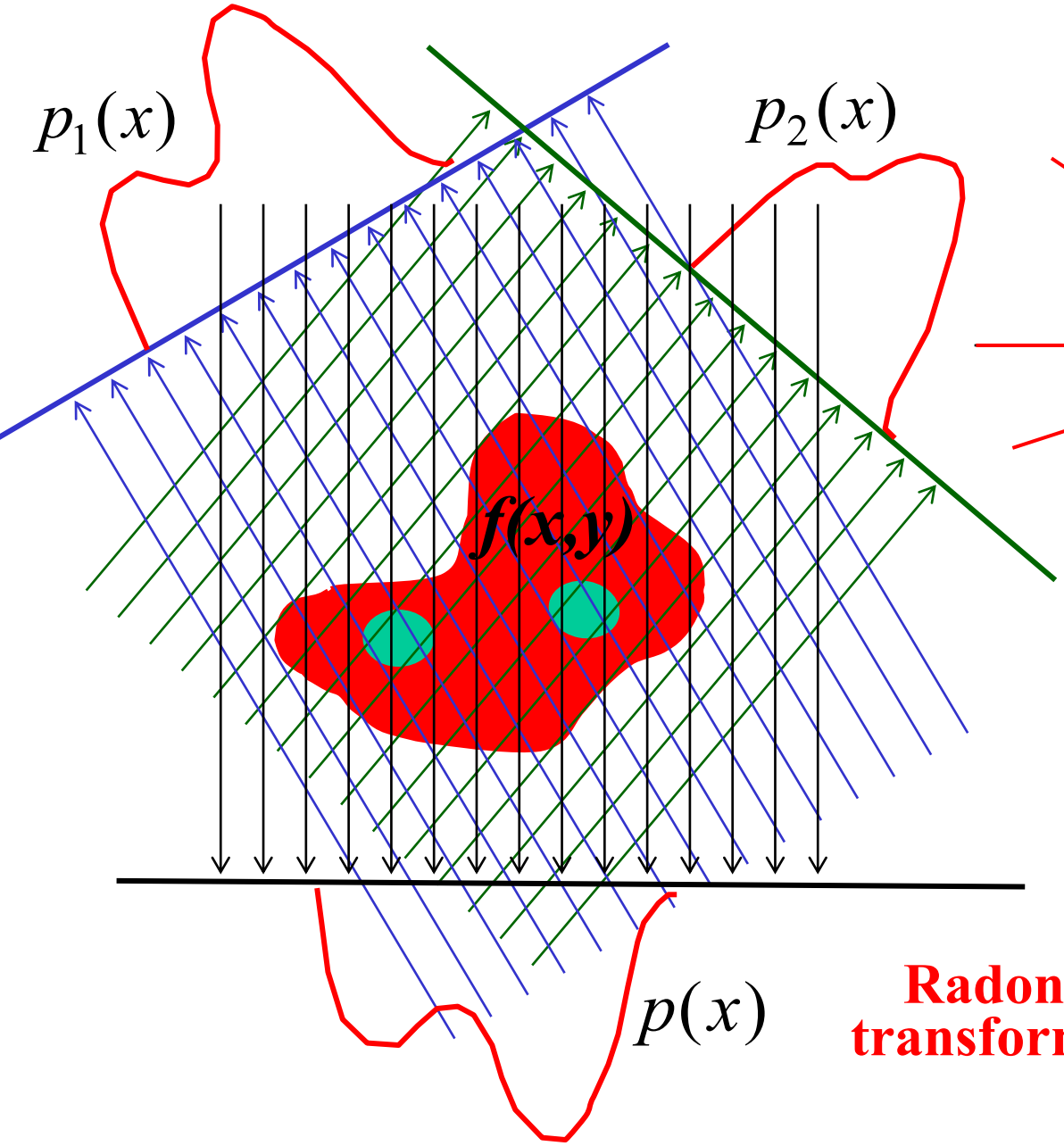
Laplacian ~ Difference of Gaussians

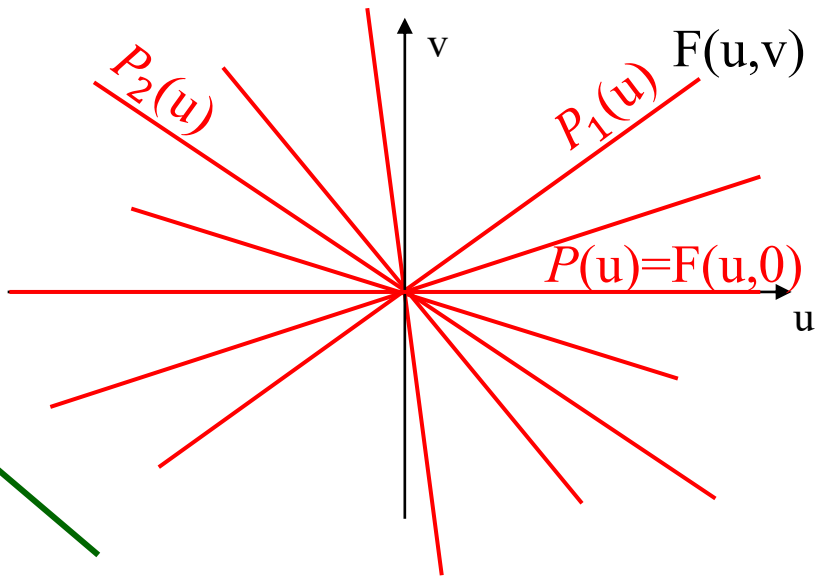


DOG = Difference of Gaussians

More details on Gaussian and Laplacian pyramids can be found in the paper by Burt and Adelson (link will appear on the website).

Computerized Tomography (CT)

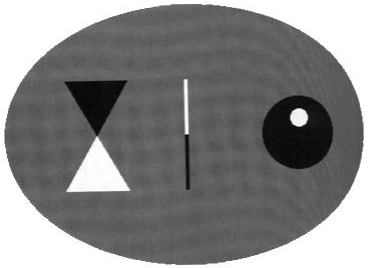




$$P(u) = F(u,0)$$

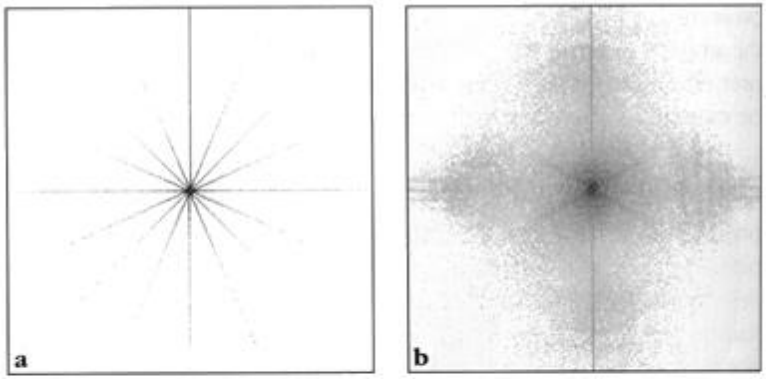
Radon transform: $$p(x) = \int_y f(x,y)dy$$

Computerized Tomography (CT)



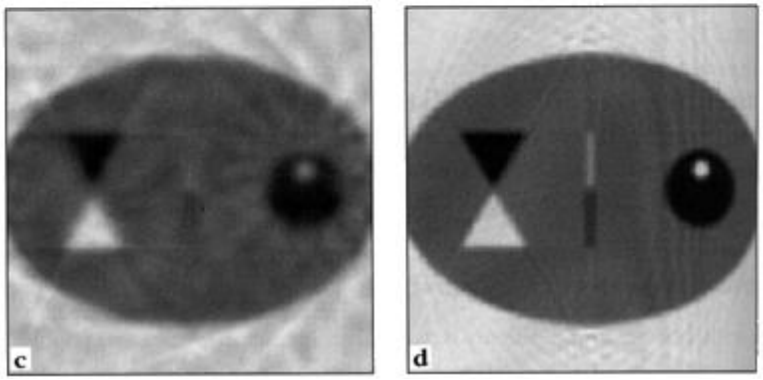
Original (simulated) 2D image

8 projections- Frequency Domain



120 projections- Frequency Domain

Reconstruction from 8 projections



Reconstruction from 120 projections

End of Lesson...

Exercise#1 -- will be posted on the website.
(Theoretical exercise: To be done and submitted individually)

To get course announcements:
➔ please sign in to the course website! (Moodle)

Vision & AI Seminar (not for credit):
Thursdays at 12:15-13:15
(in person – in Ziskind-1)